Helios

How does a small rocket fly, and bring itself back?

A 38 mm motor burns two grains of potassium nitrate and sorbitol for 1.3 s and throws the 1.15 kg rocket off a 1.2 m rail. It coasts to about 800 m. There a small computer in the main body sees the air pressure start to rise, turns a servo in the nose, and two springs kick a side door open so the parachute can come out.

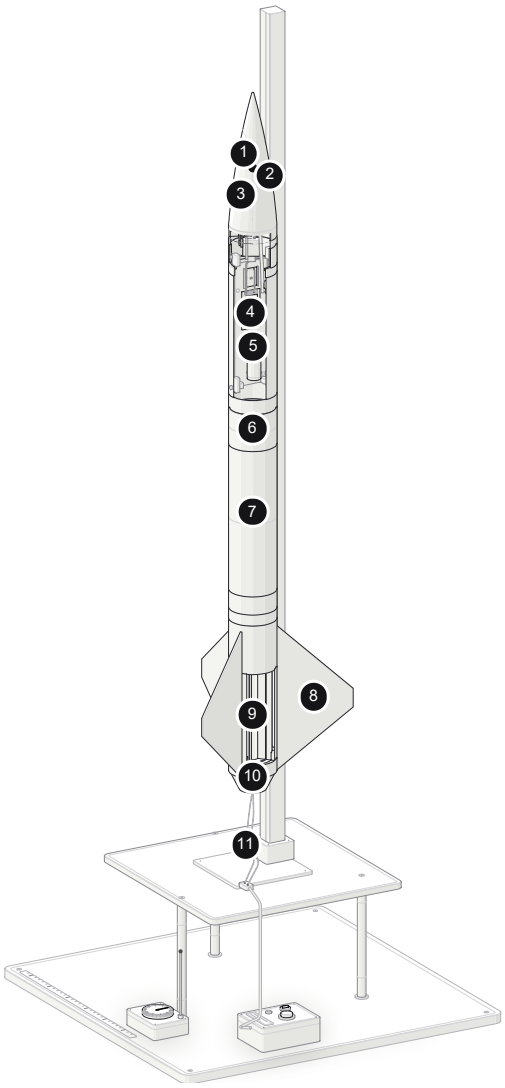
218 N·s
TOTAL IMPULSE

170 N
PEAK THRUST

800 m
APOGEE

1.15 kg
AT LIFTOFF

6.5 m/s
UNDER THE CANOPY



01 **Side-door nose**
Holds the servo latch, the springs and the packed parachute.

02 **Door**
One quarter of the nose; its foot is held by the servo's arm.

03 **Parachute**
Packed in a bag on a ramp; slides out once the door is gone.

04 **Avionics bay**
ESP32, BMP280 barometer and an 18650 cell on a sliding deck.

05 **Main body**
Anchors the Kevlar shock cord and carries the bay.

06 **Coupler sleeve**
Joins the main body to the lower body on their spigots.

07 **Lower body**
The middle tube, between the coupler and the fin can.

08 **Fin can**
Three fins through the wall, around the motor tube.

09 **38 mm KNSB motor**
Two BATES grains burning from the core and the ends.

10 **Boattail motor cap**
Holds the motor in and tapers the tail.

11 **Launch pad and rail**
A 1010 rail, a blast plate, the controller and ground station.